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Comparative Analysis of Model Performance on CLAHE Enhanced Synthetic GAN Images vs Real Fundus Images

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Abstract - Deep learning-based automated models have been of much interest for detecting many eye disorders from retinal fundus images. But the performance of their models is normally constrained by access to big, well-labeled, and diversified datasets. Data augmentation processes, including rotation, flipping, and contrast, can only create restricted variations and can fail to completely represent real-world pathological examples in their diversity. This project overcomes the data shortage issue by utilizing Generative Adversarial Networks (GANs) to generate realistic eye fundus images, which increases dataset diversity and enhances model generalization. The dataset is made up of 4,000 grayscale fundus images from the APTOS dataset, preprocessed with Contrast Limited Adaptive Histogram Equalization (CLAHE) for improved contrast. A GAN model is trained to produce synthetic images of various stages of eye diseases, which are classified into "No Disease" and "Affected Stages." CycleGAN is also used to produce additional variations with medical relevance. These synthetic images are incorporated into the training process of two deep learning models: (1) a five-layer convolutional custom Convolutional Neural Network (CNN) and (2) a transfer learning-based model based on EfficientNet-B3. To evaluate the efficacy of synthetic data, both models are trained on and without GAN-generated images. Performance is measured in terms of accuracy, F1-score, and other appropriate metrics. The results show that the addition of synthetic images enhances classification accuracy, showing the utility of GAN-generated data in medical imaging tasks. This work demonstrates the promise of GAN-based data augmentation for improving deep learning models in ophthalmology. Future research will aim to optimize GAN architectures, extend synthesis to other retinal diseases, and incorporate multimodal clinical data for more complete disease diagnosis.

Keywords: GAN; CycleGAN; Diabetic retinopathy; CLAHE; EfficientNet-B3; Retinal fundus image.

I. INTRODUCTION

Eye conditions like glaucoma, diabetic retinopathy, and age-related macular degeneration (AMD) are the global leading causes of blindness and vision loss. The conditions develop in the initial stages silently, and thus early detection and prompt treatment should be conducted in an attempt to avoid permanent damage. Conventional diagnosis utilizes skilled ophthalmologists to examine retinal fundus images for disease patterns. Manually, it is time-consuming, susceptible to human error, and unsuitable for mass screening. In the last few years, deep learning models in the form of Convolutional Neural Networks (CNNs) have demonstrated potential in automatic eye disease diagnosis and classification from fundus images. Though successful, the models need large quantities of diverse, well-labeled datasets in an attempt to be highly accurate and resilient when used in the real world. One of the largest limitations to training deep learning models to analyze medical images is that large quantities of well-labeled datasets are not available.

Datasets that are publicly available, such as APTOS and EyePACs, contain thousands of retinal images but often suffer from class imbalances. This means that certain stages of a disease may have significantly fewer examples compared to others. Additionally, variations in imaging equipment, patient demographics, and environmental conditions can introduce noise and variability, making it difficult for deep learning models to generalize effectively. Traditional data augmentation techniques, like flipping, rotating, adjusting brightness, and enhancing contrast, can artificially expand datasets but fail to introduce new pathological variations. This limitation restricts the model's ability to learn complex disease patterns, ultimately affecting its performance on unseen data. To address these challenges, Generative Adversarial Networks (GANs) offer a promising approach by generating realistic eye fundus images that closely resemble actual medical data. GANs consist of two competing neural networks: a generator that creates synthetic images and a discriminator that assesses their authenticity.

Through this adversarial method, GANs are able to produce high quality images capturing the different pathological features available in a multitude of eye disorders. In this work, we work with GANs and CycleGANs to develop fundus images with the aim of augmenting the variability of the dataset and improving the performance deep learning models. In this project, there are 4,000 gray-scale fundus pictures from the APTOS dataset, which were previously processed by Contrast Limited Adaptive Histogram Equalization (CLAHE) for enhancement of the features. A GAN model is developed to create synthetic fundus images depicting different degrees of the disease and they are labels as “No Disease” and “Affected Stages”. Additional medical relevant variations are created with a CycleGAN. These images are utilized together with other two deep learning models in the training pipeline:

1. A five convolutional layer custom-built CNN designed from scratch for classification and feature extraction is used.

2. An EfficientNet-B3 transfer learning model where features are used to aid in increasing accuracy and efficiency of the model.

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To evaluate the impact of synthetic data, the two models are trained both with and without the inclusion of GAN synthesized images. Comparison is made based on standard measures, such as accuracy, precision, recall, and further metrics like the Fréchet Inception Distance, structural similarity index and others within the domain of image quality to assess the authenticity of created images. This analysis seeks to confirm whether the use of GAN synthesized data facilitates enhanced model performance and consequently provides a robust and versatile AI-driven ophthalmic diagnostic system.

This advancement in ophthalmology underscores the role of GAN-augmented data introduced through this research in improving dataset diversity and minimizing reliance on human manual annotation.

Further research might involve moving GAN architecture optimization further, generating synthetic images of other retinal lesions, and adding multimodal clinical data (i.e. Optical Coherence Tomography, clinical file) to support a more sophisticated diagnostic methodology.

The outcome of this project could lead to more accurate, efficient, and easily deployed mechanisms for identifying eye diseases using AI technology, which will facilititate improved healthcare for patients as well as practitioners the world over.

II. LITERATURE SURVEY

Literature survey section involves speaking about Almahairi et al.. [2018] present the Augmented CycleGAN, designed for many-to-many mappings in unpaired data, which holds promise for medical image synthesis. Saeed et al. .[2021] conduct a systematic review on the use of GANs for glaucoma detection, showcasing their effectiveness in medical imaging. You et al.[2022] provide a survey of GAN applications in ophthalmology, focusing on synthetic data generation and disease classification. Tang et al. [2024] enhance CycleGAN for low-light image improvement, which is particularly useful in medical imaging where lighting conditions can impact analysis.

Raj et al. [2022] introduce a new technique to enhance fundus images, leading to improved disease detection through better clarity and contrast. Mudjirahardjo [2024] examines the impact of grayscale conversion, CLAHE, and filtering in the convolution process, which is essential for preprocessing fundus images in deep learning applications.

Koonce [2021] offers a comprehensive overview of ResNet-50, a popular deep CNN architecture. Sarvamangala and Kulkarni [2022] review techniques for medical image understanding based on CNNs, including classification and segmentation. Sinha and El-Sharkawy [2019] propose Thin MobileNet, an optimized version of MobileNet aimed at efficient image processing. Freeman et al. [2018] introduce Effnet, an efficient CNN architecture that improves image classification accuracy while minimizing computational costs. Li et al. [2014] investigate CNNs for medical image classification, marking one of the earliest explorations of deep learning in healthcare.

Zhuang et al. [2019] discuss various transfer learning strategies and their applications in medical imaging. Weiss et al. [2016] provide a foundational survey on transfer learning, highlighting its influence on deep learning models in small medical datasets.

Jordon et al. [2018] explore the importance of synthetic data in machine learning, detailing the reasons and methods for its generation. Kovalyk et al. [2024] introduce the PAPILA dataset, which comprises fundus images and clinical data for glaucoma assessment, enhancing the availability of high-quality datasets for research.

Mahendran et al. [2021] employ a Deep Learning-Based Adaptive Recurrent Neural Network for myocardial infarction diagnosis, which establishes the efficacy of deep learning models in medical diagnosis. Their method enhances cardiac image analysis using adaptive learning.SenthilPandi et al. [2023] introduce hyper-parameter optimization of machine learning algorithms to enhance classification accuracy. Their work is useful in the area of medical imaging, where hyperparameter tuning can enhance model performance substantially.

Jabbar et al. [2022] provide a concise introduction to Generative Adversarial Networks (GANs) by separating their variants, limitations, and applications into various sections. It is applicable in the context of GAN's use in medical image generation and data augmentation. Wang and Lin [2024] enhance CycleGAN by optimizing its cyclic consistency, which leads to better unpaired medical image translation performance. The contribution can be applied in domain adaptation and image enhancement of medical images.

Bellemo et al. [2019] use GANs to synthesize retinal fundus images, proving the potential of synthetic data generation to augment disease classification model training sets, particularly in ophthalmology.Sandfort et al. [2019] use CycleGAN to augment data in CT segmentation tasks for improving model generalizability. Their study proves the potential of GAN-based augmentation in medical imaging tasks.

Yu et al. [2022]give an overview of transfer learning methods to medical image analysis, attributing their effectiveness in handling small annotated sets. From their findings, transfer learning is an effective solution to medical image classification issues.Haddadi et al. [2024] introduce a new medical image improvement algorithm through the combination of CLAHE and pelican optimization. Their algorithm improves contrast and details of medical images for better disease diagnosis and analysis.

Xia et al. [2022] summarize GAN inversion techniques, whose medical imaging applications are reconstructing and editing medical images for better interpretability.Liu et al. [2024] present a CycleGAN-based approach to virtually staining freshly cut brain tissue for better visualization without employing traditional staining techniques. Their research demonstrates the capability of GANs in histopathological examination.

Almazroa et al. [2018] present the RIGA dataset, a public dataset for glaucoma analysis from retinal fundus images. It is a useful resource for the training of deep learning models for ophthalmology.Kiefer et al. [2023] present a repository of public datasets for glaucoma, presenting a complete resource for machine learning-based glaucoma detection using retinal fundus and OCT images.Rifa'i et al. [2024] explore the diagnosis of pneumonia based on MobileNetV2 for chest X-rays with enhanced image techniques such as white balance and CLAHE. They highlight the contributions of deep learning in the automation of disease diagnosis.

The SPIE 10579 proceedings[2018] offer various applications of medical imaging in healthcare, research, and artificial intelligence that aid in enhancing imaging informatics as well as automated diagnosis. Setiawan et al. [2013] introduce a CLAHE-based technique for color retinal image

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improvement, enhancing the quality of images for improved visualization of retinal scan abnormalities and blood vessels.

The research paper by Farjana, Senthilnaathan, and Rohit [2025] introduces the generation of synthetic medical images through Generative Adversarial Networks (GANs) and Contrast Limited Adaptive Histogram Equalization (CLAHE) processing. The research aims to improve the quality of synthetic images and compare the performance of different deep learning models on real and synthetic data. The research emphasizes the efficiency of CLAHE in improving contrast in images, resulting in improved synthetic data generation. Comparison of different models is done to determine the effect of synthetic data on classification performance, especially in medical imaging tasks. The research emphasizes the possibility of training deep learning models with synthetic images, showing that well-generated synthetic data can improve model generalization and solve the problem of data scarcity in medical diagnosis.

III. EXISTING SYSTEM

Present deep learning-based methods for eye disease diagnosis are dependent on publicly available ophthalmologist-labeled retinal fundus image datasets. These datasets are used to train convolutional neural networks (CNNs) to classify various stages of diseases. One of the biggest challenges is that these datasets are small in size and imbalanced. Some disease phases have fewer samples, resulting in over-representative models that have high accuracy for prevalent conditions but are poor at predicting rare ones. Such imbalanced effects impact the generalization capability of deep learning models, rendering them less valuable in actual clinical applications. To make up for the insufficient data, the common data augmentation methods like rotation, flipping, brightness change, and contrast enhancement are used. These operations do add minor changes but do not generate new pathological features that enable the model to learn more diverse representations. The model can, therefore, still overfit certain dataset features and be unable to generalize to unknown images from alternative populations or imaging equipment.

Another drawback of the current system is its reliance on supervised learning, which is a data-hungry and manual process for annotating the data. Nevertheless, expert annotation is costly, labor-intensive, and error-prone. Moreover, datasets often contain mislabeled or partially labeled data, further decreasing the robustness of learned models. Also, deep models learned with limited datasets have difficulties when exposed to images coming from other sources, thus limiting their real-world applications.

Even with the progress in medical imaging, synthetic data usage is still underexploited in ophthalmology. Generative Adversarial Networks (GANs) can potentially produce high-quality synthetic images that can add new pathological variations, but current systems do not incorporate them into the training process to a large extent. A stronger method is required to increase dataset diversity and enhance the generalization capability of deep learning models for eye disease classification.

IV. PROPOSED SYSTEM

The hybrid approach of this system fills in the gaps of other existing methods by integrating the strengths of GANs and VAEs to create a high-quality synthetic image for the fundus that could be used in classifying various ocular diseases such as Age-related Macular Degeneration (ARMD), glaucoma, diabetic retinopathy, and cataracts. A hybrid generative architecture of the proposed framework allows effective synthesis of fundus images that capture the variability in real-world medical data.

It starts by gathering the dataset by getting images of the fundus from sources such as HRF and Kaggle, that is, both normal and pathological conditions like glaucoma, ARMD, cataracts, and diabetic retinopathy. The image pre-processing technique follows by resizing the images into a standard size and further enhancing them with CLAHE to improve clarity and bring out critical features in GAN training.

For an overall evaluation of performance using CNN architectures based on ResNet and MobileNet are evaluated for classification performance. Three different cases, one on real images and the other on synthetics, and the third, to be used for benefit, on a mixed set containing both real and synthetically generated images, each using a CNN model under each of these cases is conducted. This architecture seeks the high precision of CNN's classification of ocular disease while addressing the problem presented by small medical datasets with inherent limitations.
Figure 1 represents the model creation workflow

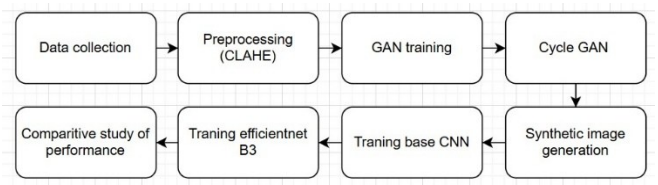

Fig.1. Flow Diagram

V. METHEDOLOGY

A. Dataset Description

This project utilizes various datasets based on different classes of eye fundus needs, eyePACS a dataset that is recognized widely for analysis of glaucoma, PAPILA for healthy eyes, ARIA for age related macular detection, HRF Datasets for combination of all classes in all. Finally FIVES Dataset is used for testing of all data.

A large amount of eye fundus images that represents various eye classes. However, for the purpose of the model, narrowing the dataset to have four specific classes: glaucoma, diabetic retinopathy, cataract, ARMD and healthy eyes. These classes were picked based upon the clinical prevalence and importance.

Table 1 represents the datasets used and count of image data available in each of them.

TABLE 1

List of all datasets with counts

Datasets	Images count
Aptos-2019	3662

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B. System Design and Implementation

The system is developed a research project which focusses on study of accuracy of synthetic images vs real images in a CNN based model, the synthetic images are generated by feeding the CLAHE processed images to Normal and CYCLE GAN architecture that focusses which generates image and its pushed to the generator of GAN to which fools the generator to produce synthetic fake images similar to input data.

Next we focus on developing a CNN architecture model based on efficientnet b3 and base CNN model too. Train the models based on 3 categories – 1. Real data, 2. Synthetic data, 3. Mixed data (combination) and optimization techniques are applied to fetch the accuracy on each case which is then studied and compare across various models.

C. Requirement Specifications

The requirement specification for skin disease classification and care recommendation based on deep learning gives a comprehensive roadmap to the components that are required to design and create a diagnostic tool. The key purpose of requirement analysis is to give a complete view of the elements that is necessary to build an accurate and reliable system that aims at detection at the earliest and care guidance for skin diseases.

1) Libraries and Frameworks:

- OpenCV: Image processing task is done by OpenCV that includes rescaling image to the needed resolution, also performs data augmentations like zoom, translation, rotation and flipping that improves the performance and robustness of the model.

- Pandas and NumPy: Pandas is used for manipulation of data and data structure management for example Data Frames allowing smooth data handling all through the project. NumPy is employed for numerical calculation

- Keras: Keras a high-level neural networks API that is used for building and training the CNN (Convolution Neural Network). The backend used by keras is TensorFlow

- TensorFlow: Easy CNN creation for skin disease prediction is made possible by TensorFlow's tools such as keras that optimizes computational graphs for speedy training and GPU interoperability

- · NVIDIA CUDA and cudNN : Nvidia drivers used for activation of GPU for utilizing GPU for processing and training the various models like GAN and CNN with tensorflow

- PyTorch is an open-source machine learning library developed by Facebook's AI Research lab (FAIR), widely used for deep learning applications.PyTorch supports tensor computations with GPU acceleration, making it highly efficient for training large neural networks. It includes a rich ecosystem of tools and libraries, such as torchvision for computer vision tasks, torchaudio for audio processing, and torchtext for natural language processing. PyTorch's intuitive interface, combined with its strong community support and seamless integration with Python.

2) Hardware requirements :

- Nvidia RTX 3050 or more: Complex calculations, and sophisticated algorithms

- Graphics Processing Units (GPUs). Notably, GPUs are better for parallel computations; therefore, they are beneficial in machine learning and scientific applications.
- Memory and Storage: For temporary storage of data while computing, one requires RAM, which is basically called Random Access Memory. One needs SSDs and HDDs for persistent data storage. High-speed and large-capacity storage solutions required .

VI.MODULE DESIGN

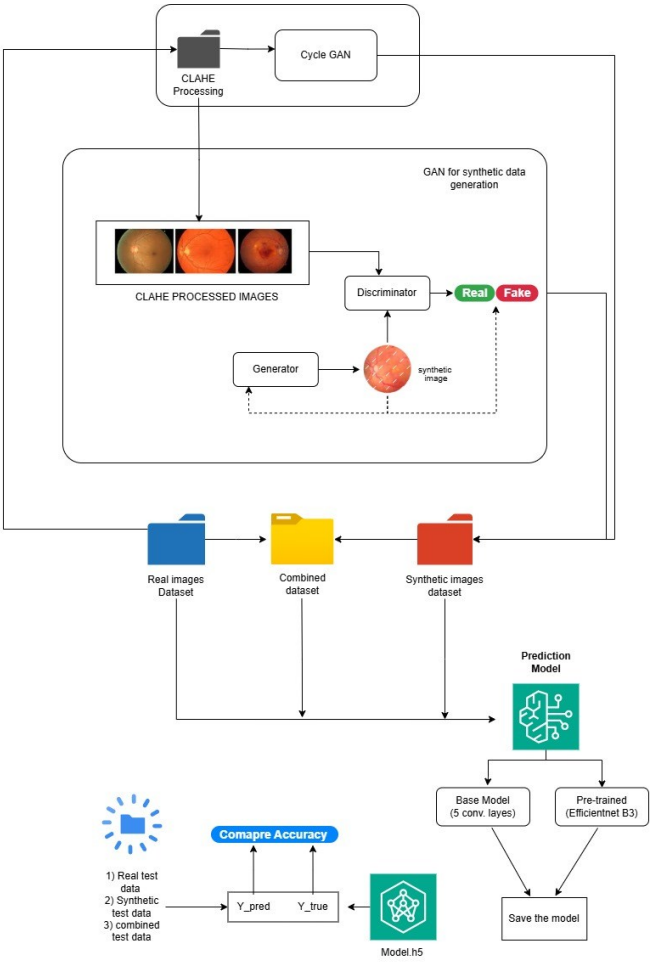

Fig. 3. Architecture Diagram

1. Data Collection Module

There are various work stages, starting from massive collection of different types of fundus images through the credible High-Resolution Fundus image database and Kaggle, which include a broad variety of images varying from the above descriptions: eye diseases such as glaucoma, age-related macular degeneration, cataracts, diabetic retinopathy, and relatively healthy eye images. This means pretty much aim at developing a challenging and diverse data set to better improve the training and test processes over the models. When each picture is selected, there is care in choosing high-quality images relevant to the ensuing stages of preprocessing, data augmentation, and then training the model. To get a sufficiently representative sample accounting for the presence

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of considerable variability in ocular conditions, which would be pivotal in the building of models that generalize well across many patient populations.

1. *Data Augmentation using CLAHE*

Once collected, the images are to be preprocessed and enhanced, in terms of quality, in order to have them standardized in their dimensions for use in GANs and other deep learning models. Each of these images is required to be resized to a uniform size. Subsequently, the images were passed through Contrast Limited Adaptive Histogram Equalization (CLAHE) to improve the contrast of the images. CLAHE will redistribute the light intensity in the images, so features that are important may have been less noticeable in the original images. The benefit of this enhancement for medical images is that the models can learn more clearly the critical features. It further ensures that the images undergo preprocessing to ensure that the images are in their optimal condition for training.

2. *Training GAN for Data Generation*

The process of creating synthetic images in this method requires training a Generative Adversarial Network (GAN) to produce realistic synthetic fundus images. The GAN includes two neural networks: a generator, into which random noise is inputted and that is trained to produce realistic fundus images, and a discriminator, through which the images produced are estimated and separated between real and synthesized. By using an iterative adversarial training strategy, in which the generator progressively enhances its capacity to generate realistic images while the discriminator improves its capacity to identify real from synthetic, the GAN progressively refines its 64 x 64 output. This ensures that the generated images retain vital features and patterns of eye disease, such that they are useful for data augmentation and training deep learning models.

3. *CycleGAN Training Module*

The **CycleGAN Training Module** generates synthetic fundus images by learning the mapping between "No Disease" and "Stages 1-4" domains. It uses two generators (G_A2B and G_B2A) and two discriminators, with cycle-consistency loss ensuring image integrity during translation. The training alternates between adversarial loss (distinguishing real/fake images) and cycle-consistency loss. After training, synthetic images are created and combined with the original dataset to augment CNN training. Implemented using PyTorch, CycleGAN is ideal for medical datasets as it doesn't require paired images and produces realistic, diverse data. These synthetic images are integrated into the Dataset Preparation Module and evaluated for their impact on glaucoma detection model accuracy.

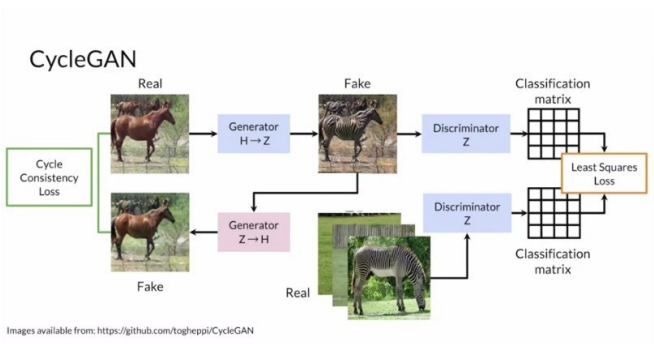

Fig 4. Representation of CycleGANs

4. *Comparative Study of CNN Models with Real, Synthetic, and Mixed Data*

Three The final phase of the project is the detailed assessment of synthetic impact in the prediction of diseases through three different architectures of a Convolutional Neural Network: Effnet B3 and a custom CNN model. Each of the CNNs is trained on three different datasets: one on real images, one on synthetic images, and one on a mixed dataset comprising both real and synthetic images. Performance of these models is tested strictly using a set of different metrics, for example, accuracy, precision, recall, and an F1 score. It is a comparative study which provides a result for evaluation to assess how well the models trained on synthetic and mixed data will perform in comparison to those which are trained solely on real data. By analyzing the outcome, try to see if synthetic data is significant for augmenting the training process and improving model generalization toward ultimately being helpful for the accurate prediction and diagnosis of eye diseases. The results from this study will provide meaningful insights into the possibility of GAN-generated data in medical imaging and promote the field of deep learning in disease detection and diagnosis. Findings can be used to explain and justify the replacement of real data with synthetic data as a complementary component that may result in better performance and diagnosis in clinical environments.

VII RESULT AND DISCUSSION

Data Collection: A diverse dataset comprising fundus images of various eye diseases (glaucoma, ARMD, cataract, diabetic retinopathy, and healthy eyes) was meticulously curated from reputable sources HRF, Aptos.etc.

Image Preprocessing: To enhance image quality and facilitate model training, Contrast Limited Adaptive Histogram Equalization (CLAHE) was applied to all images, improving contrast and detail visibility. The data is streamlined into a folder and a better image with better contrast is obtained for further analysis of images. Then the resultant images are stored in a folder for future processing.

Ordinary histogram equalization produces an image with higher global contrast. CLAHE improves on this by calculating multiple histograms for different sections of the image, and then redistributing the brightness values.

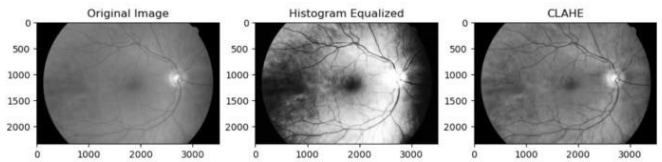

Fig.5. CLAHE Processing

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CLAHE-GAN: A GAN model was trained on CLAHE-processed images to assess the impact of enhanced image quality on synthetic image generation. These CLAHE images have a better contrast as compared real images highlighting the necessary parts of image in detail but limited to clip limit. GAN was trained for 8000 EPOCHS and the following results are streamlined to store themselves in a folder for further image generation process. Reaching SSIM of 0.59, min of 0.08 to max of 0.29 LPIPS score with 64 x 64 images.

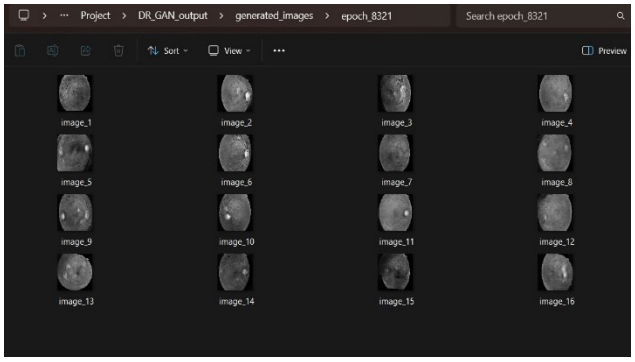

Fig 6. GAN Generated images at 8321th epoch

CYCLE-GAN: A PyTorch based implementation of GAN with 2 discriminators and 2 generators instead of one. The input image from dataset is fed and it generates combinatory images – Real, Fake and reconstructed. i.e., 4 images per epochs. Ran this for 52 epochs and got 200+ synthetic images attached below. Reaching SSIM of 0.89 and around 0.4 of LPIPS score.

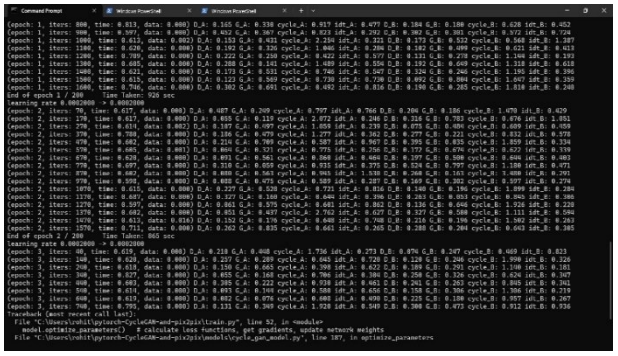

Fig.7. Training the CycleGAN images

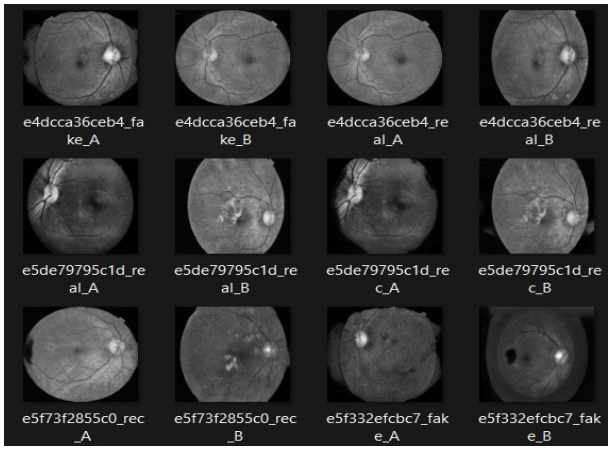

Fig. 8. Generated images of Cycle GAN

CNN model prediction:

In this study, we evaluated the performance of EfficientNet-B3 for glaucoma detection using two validation strategies: K-Fold Cross-Validation and Standard Train-Validation Split. The results demonstrated the effectiveness of EfficientNet-B3 in classifying diabetic retinopathy and non-disease cases. Achieved accuracy of 89% with 64 x 64 limited quality images while the Cycle GAN images produced accuracy of 80% and a effnet-b3 based cnn approach produces around 82% accuracy which is pretty good one for 64 x 64 images produced.

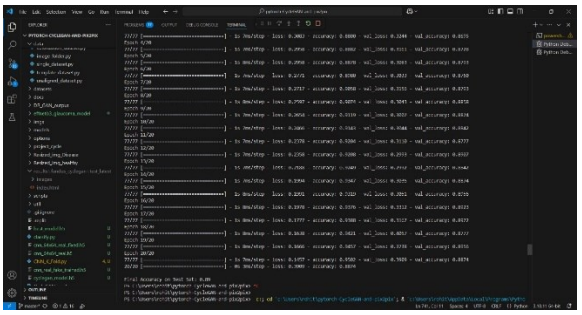

Fig. 9. Normal train test split accuracy (real 64 x 64)

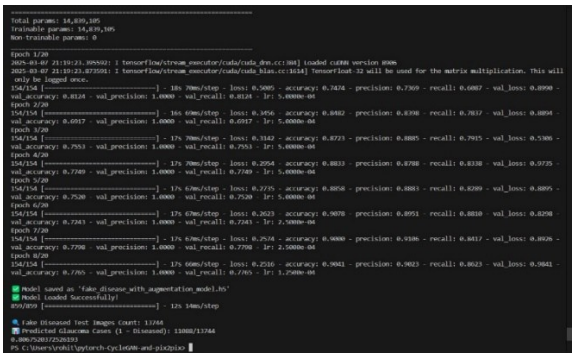

Fig 10 Fake images 64 x 64 tested against Effnet B3

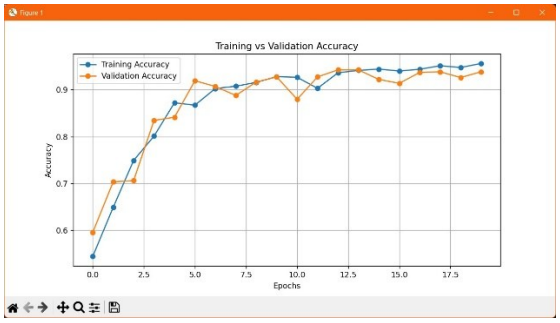

Fig 12 – model performance on testing and validation data.

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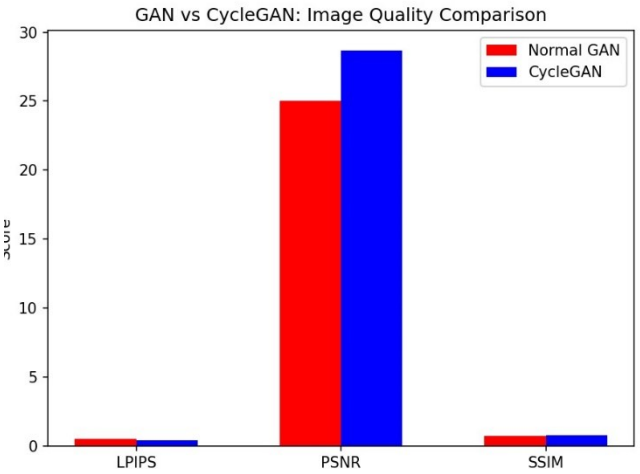

Fig 12 comparison of generated images on metrics (LPIPS, PSNR, SSIM)

VII CONCLUSION AND FUTURE SCOPE

This work illustrates the performance of Generative Adversarial Networks (GANs) to create synthetic eye fundus images for improving deep learning models to classify eye disease. Through resolving data sparsity and class imbalance, GAN-synthesized images yield a more varied dataset, enhancing generalization of the model. Two deep learning models—a bespoke CNN and a transfer learning-based EfficientNet-B3—were trained and tested with and without the synthetic data. The outcomes reveal that the integration of synthetic images enhances classification performance, which supports the capability of GANs in medical image augmentation. These observations endorse the efficacy of synthetic data to address dataset issues and increase the robustness of models in practical ophthalmic use.

Subsequent research can concentrate on improving GAN models so that they can produce even higher-resolution and more realistic fundus images with higher pathological details. Moreover, extending the synthesis method to other conditions like diabetic retinopathy and age-related macular degeneration (AMD) can increase its potential. Combining multimodal data like Optical Coherence Tomography (OCT) scans and patient history can enhance diagnostic precision further. Additionally, investigation of semi-supervised and self-supervised learning methods in addition to synthetic data may lower reliance on human-labeled datasets. Finally, applying the trained models to the real world and assessing their real-world effectiveness would be an important step toward creating an automated and trustworthy eye disease screening system.

REFERENCES

[1] A. Almahairi, S. Rajeshwar, A. Sordoni, P. Bachman, and A. Courville, "Augmented CycleGAN: Learning Many-to-Many Mappings from Unpaired Data," in *Proc. 35th Int. Conf. Mach. Learn.*, vol. 80, pp. 195–204, 2018.

[2] A. Raj, N. A. Shah, and A. K. Tiwari, "A novel approach for fundus image enhancement," *Biomed. Signal Process. Control*, vol. 71, no. Part B, p. 103208, 2022, doi: 10.1016/j.bspc.2021.103208.

[3] A. Saeed, S. N. Sheikh Abdullah, J. Che-Hamzah, and A. Abdul Ghani, "Accuracy of Using Generative Adversarial Networks for Glaucoma Detection: Systematic Review and Bibliometric Analysis," *J. Med. Internet Res.*, vol. 23, no. 5, 2021.

[4] A. You, J. K. Kim, I. H. Ryu et al., "Application of generative adversarial networks (GAN) for ophthalmology image domains: a survey," *Eye Vis.*, vol. 9, no. 6, 2022.

[5] B. Koonce, "ResNet 50," in *Convolutional Neural Networks with Swift for Tensorflow*. Berkeley, CA: Apress, 2021, doi: 10.1007/978-1-4842-6168-2_6.

[6] D. R. Sarvamangala and R. V. Kulkarni, "Convolutional neural networks in medical image understanding: a survey," *Evol. Intel.*, vol. 15, pp. 1–22, 2022, doi: 10.1007/s12065-020-00540-3.

[7] D. Sinha and M. El-Sharkawy, "Thin MobileNet: An Enhanced MobileNet Architecture," in *2019 IEEE 10th Annu. Ubiquitous Comput. Electron. Mobile Commun. Conf. (UEMCON)*, New York, NY, USA, 2019, pp. 0280–0285, doi: 10.1109/UEMCON47517.2019.8993089.

[8] F. Zhuang et al., "A Comprehensive Survey on Transfer Learning," *Proc. IEEE*, vol. 109, no. 1, pp. 43–76, 2019.

[9] G. Tang et al., "An Improved CycleGAN-Based Model for Low-Light Image Enhancement," *IEEE Sens. J.*, vol. 24, pp. 21879–21892, 2024.

[10] I. Freeman, L. Roese-Koerner, and A. Kummert, "Effnet: An Efficient Structure for Convolutional Neural Networks," in *2018 25th IEEE Int. Conf. Image Process. (ICIP)*, Athens, Greece, 2018, pp. 6–10, doi: 10.1109/ICIP.2018.8451339.

[11] J. Jordon et al., "Synthetic Data—What, Why and How?," *arXiv preprint arXiv:2205.03257*, 2022, doi: 10.48550/arXiv.2205.03257.

[12] K. Weiss, T. M. Khoshgoftaar, and D. Wang, "A survey of transfer learning," *J. Big Data*, vol. 3, no. 9, 2016, doi: 10.1186/s40537-016-0043-6.

[13] O. Kovalyk et al., "PAPILA: Dataset with fundus images and clinical data of both eyes of the same patient for glaucoma assessment," *Sci. Data*, vol. 11, no. 1, p. 391, 2024, doi: 10.1038/s41597-024-03175-6.

[14] P. Mudjirahardjo, "The Effect of Grayscale, CLAHE Image and Filter Images in Convolution Process," *Int. J. Adv. Multidiscip. Res. Stud.*, vol. 4, no. 3, pp. 943–946, 2024.

[15] Q. Li et al., "Medical image classification with convolutional neural network," in *2014 13th Int. Conf. Control Autom. Robot. Vis. (ICARCV)*, Singapore, 2014, pp. 844–848, doi: 10.1109/ICARCV.2014.7064414.

[16] R. K. Mahendran et al., "Deep Learning Based Adaptive Recurrent Neural Network for Detection of Myocardial Infarction," *J. Med. Imaging Health Inform.*, vol. 11, no. 12, pp. 3044–3053, 2021, doi: 10.1166/jmihi.2021.3913.

[17] S. S. Pandi, V. R. Chiranjeevi, K. T, and P. Kumar, "Improvement of Classification Accuracy in Machine Learning Algorithm by Hyper-Parameter Optimization," in *2023 Int. Conf. Res. Methodol. Knowl. Manag. Artif. Intell. Telecommun. Eng. (RMKMATE)*, Chennai, India, 2023, pp. 1–5, doi: 10.1109/RMKMATE59243.2023.10369177.

[18] S. Jabbar et al., "Generative Adversarial Networks (GANs): Introduction, Taxonomy, Variants, Limitations, and Applications," *Multimed. Tools Appl.*, vol. 81, no. 4, pp. 1–27, 2022.

[19] T. Wang and Y. Lin, "CycleGAN with Better Cycles," *arXiv preprint arXiv:2408.15374*, 2024.

[20] V. Bellemo et al., "Generative Adversarial Networks (GANs) for Retinal Fundus Image Synthesis," in *Comput. Vis. – ACCV 2018 Workshops*, G. Carneiro and S. You, Eds. Springer, 2019.

[21] V. Sandfort, K. Yan, P. J. Pickhardt et al., "Data augmentation using generative adversarial networks (CycleGAN) to improve generalizability in CT segmentation tasks," *Sci. Rep.*, vol. 9, no. 16884, 2019, doi: 10.1038/s41598-019-52737-x.

[22] X. Yu et al., "Transfer learning for medical images analyses: A survey," *Neurocomputing*, vol. 489, pp. 230–254, 2022, doi: 10.1016/j.neucom.2021.08.159.

[23] Y. Haddadi, B. Mansouri, and F. Z. D. Khodja, "A novel medical image enhancement algorithm based on CLAHE and pelican optimization," *Multimed. Tools Appl.*, vol. 83, pp. 90069–90088, 2024.

[24] Y. Xia et al., "A Survey on GAN Inversion: Techniques and Applications," *ACM Comput. Surv.*, 2022.

[25] Z. Liu et al., "Virtual formalin-fixed and paraffin-embedded staining of fresh brain tissue via stimulated Raman CycleGAN model," *Sci. Adv.*, vol. 10, 2024.

[26] A. Almazroa et al., "Retinal fundus images for glaucoma analysis: the RIGA dataset," in *Proc. SPIE 10579, Med. Imaging 2018: Imaging Inform. Healthc., Res., Appl.*, vol. 105790B, 2018, doi: 10.1117/12.2293584.

[27] R. Kiefer et al., "A Catalog of Public Glaucoma Datasets for Machine Learning Applications: A detailed description and analysis of public glaucoma datasets available to machine learning engineers tackling glaucoma-related problems using retinal fundus images and OCT images," in *Proc. 2023 7th Int. Conf. Inf. Syst. Data Min.*, 2023.

[28] A. M. Rifa'i et al., "Analysis for diagnosis of pneumonia symptoms using chest X-ray based on MobileNetV2 models with image enhancement using white balance and contrast limited adaptive histogram equalization (CLAHE)," *Biomed. Signal Process. Control*, vol. 90, p. 105857, 2024.

[29] *Proc. SPIE 10579, Med. Imaging 2018: Imaging Inform. Healthc., Res., Appl.*, vol. 105790B, 2018, doi: 10.1117/12.2293584.

[30] A. W. Setiawan, T. R. Mengko, O. S. Santoso and A. B. Suksmono, "Color retinal image enhancement using CLAHE," International Conference on ICT for Smart Society, Jakarta, Indonesia, 2013, pp. 1-3, doi: 10.1109/ICTSS.2013.6588092. keywords: {Retina; Image colour analysis; Image enhancement; Colored noise; Biomedical imaging; Image quality; Blood vessels; color; retinal image; asymmetric; enhancement; CLAHE},

[31] U. Farjana, K. Senthilnaathan, and M. Rohit, "Generation of Synthetic Data Using GANs With CLAHE Processed Images and Comparative Study of Model Performance on Real and Synthetic Dataset," IEEE, 2024.

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